

Full Solution Packet

A single line determines all twisted spherical means on $\mathbb{C}$

Status: candidate full solution, likely valid

Target: the odd-Coxeter conjecture in arXiv:1103.4571

Source and exact conjecture

The source is R. K. Srivastava, *Coxeter system of lines are sets of injectivity for the twisted spherical means on $\mathbb{C}$* , arXiv:1103.4571 [1]. On page 14, after proving injectivity for Coxeter systems with an even number of lines, the paper asks the following.

(c). Consider Coxeter system of N -lines $\Sigma_N = \cup_{l=0}^{N-1} \{te^{i\theta_l} : \theta_l = \frac{\pi l}{N}, t \in \mathbb{R}\}$. Suppose $\theta_l = \pi/2$. Then $l = N/2$. By Theorem 3.6 it follows that any Coxeter system of even number of lines is also a set of injectivity for the TSM on $L^q(\mathbb{C})$.

Next, we set to describe the problem for any Coxeter system of odd lines. Let $1, \omega, \omega^2, \dots, \omega^{N-1}$ be the N roots of unity. Then $\Sigma_N = \cup_{l=0}^{N-1} \{\omega^l x : x \in \mathbb{R}\}$. We have formulated this problem in the following way.

Conjecture. Let $f \in L^2(\mathbb{C})$. Suppose $f \times \mu_r(\omega^l x) = 0, \forall r > 0$, and $l = 0, 1, \dots, N-1$ and $\forall x \in \mathbb{R}$. Then $f = 0$ a.e.

We would like to produce some evidence about the feasibility of this problem. In this case, we also get a system of solvable recursion relations by decomposing the series using symmetries, however those recursion relations for higher values of k , gives rise to a higher order square matrix, which is needed to show non-singular. This is the only difficulty in getting a solution to this problem. Since $Q_k(\omega^l x) = 0, \forall l; l = 0, 1, \dots, N-1$ and $\forall x \in \mathbb{R}$, as similar to Theorem 3.6 we will have the following conditions.

In the source notation, if ω is an N th root of unity and $\Sigma_N = \cup_{l=0}^{N-1} \omega^l \mathbb{R}$, the conjecture says that for $f \in L^2(\mathbb{C})$,

$$f \times \mu_r(\omega^l x) = 0 \quad (r > 0, x \in \mathbb{R}, 0 \leq l < N) \quad \implies \quad f = 0 \quad \text{a.e.}$$

The surrounding discussion treats odd N as the unresolved case. The result below proves the stronger assertion that the centers on just one line through the origin suffice, and it works for every L^p with $1 \leq p \leq 2$.

Definitions and normalization

For $z, w \in \mathbb{C}$, write

$$(f \times g)(z) = \int_{\mathbb{C}} f(z-w)g(w)e^{\frac{i}{2}\text{Im}(z\bar{w})} dw.$$

If μ_r is normalized arc measure on $|w| = r$, then the twisted spherical mean is

$$f \times \mu_r(z) = \int_{|w|=r} f(z-w)e^{\frac{i}{2}\text{Im}(z\bar{w})} d\mu_r(w).$$

Let

$$\varphi_k(z) = L_k \left(\frac{|z|^2}{2} \right) e^{-|z|^2/4}, \quad Q_k f = f \times \varphi_k.$$

Harmless nonzero normalization constants in φ_k do not affect any vanishing statement. The special Hermite operator is

$$\mathcal{L} = -\Delta + \frac{|z|^2}{4} - i \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right), \quad z = x + iy.$$

The standard special Hermite expansion says that $Q_k f$ belongs to the $\mathcal{L}$ -eigenspace with eigenvalue $2k + 1$, and the collection of these projections reconstructs f [1, 3]. We include the part of this structure needed for the new uniqueness argument.

Result

Theorem 1 (Single-line injectivity). *Let $1 \leq p \leq 2$ and $f \in L^p(\mathbb{C})$. Suppose*

$$f \times \mu_r(x) = 0 \quad \text{for every } x \in \mathbb{R} \text{ and } r > 0.$$

Then $f = 0$ almost everywhere.

Corollary 2 (Odd Coxeter systems). *Every line through the origin is a set of injectivity for twisted spherical means on $L^p(\mathbb{C})$, $1 \leq p \leq 2$. Consequently every Coxeter system Σ_N is a set of injectivity, whether N is odd or even. In particular, the conjecture on page 14 of arXiv:1103.4571 holds.*

Proof intuition

The spherical-mean hypothesis forces every spectral projection $Q_k f$ to vanish on the real axis. The missing point in the source is to show that restriction to the axis is injective on each infinite-dimensional Landau level.

Introduce annihilation and creation operators

$$A = \partial_z + \frac{\bar{z}}{4}, \quad A^\dagger = -\partial_{\bar{z}} + \frac{z}{4}.$$

Every vector in the k th level is $(A^\dagger)^k$ applied to an anti-holomorphic Fock vector $e^{-|z|^2/4} h(\bar{z})$. On the real axis, the equation that this vector vanishes is a normal-ordered differential equation

$$\sum_{j=0}^k \binom{k}{j} \left(\frac{x}{2} \right)^{k-j} (-1)^j h^{(j)}(x) = 0.$$

After $h(x) = e^{x^2/4} p(x)$, it becomes a constant-coefficient ODE $P_k(D)p = 0$, where

$$P_k(s) = 2^{-k} i^k H_k(is).$$

The zeros of the Hermite polynomial H_k are real and simple, so p is a finite sum of exponentials with distinct real frequencies. The Fock norm, however, reduces to the L^2 norm of that exponential sum along every horizontal line. Such a sum has positive mean square unless it is zero. Thus each $Q_k f$ is zero everywhere; special-Hermite reconstruction then gives $f = 0$.

The eigenspace restriction theorem

Put

$$\mathcal{E}_k = \ker_{L^2(\mathbb{C})}(\mathcal{L} - (2k+1)I).$$

Lemma 3 (Creation representation). *For every $v \in \mathcal{E}_k$ there is an entire function h such that*

$$v(z) = c_k (A^\dagger)^k \left(e^{-|z|^2/4} h(\bar{z}) \right), \quad c_k \neq 0, \quad (1)$$

and

$$\int_{\mathbb{C}} |h(\bar{z})|^2 e^{-|z|^2/2} dz < \infty. \quad (2)$$

Proof. A direct calculation gives

$$[A, A^\dagger] = \frac{1}{2}I, \quad \mathcal{L} = 4A^\dagger A + I.$$

Thus A lowers the eigenvalue by 2 and $A^\dagger$ raises it by 2. If $v \in \mathcal{E}_k$, then

$$A^\dagger A v = \frac{k}{2} v.$$

For $k > 0$ this identity can be iterated, giving

$$v = \frac{2^k}{k!} (A^\dagger)^k A^k v.$$

Set $u = A^k v$. Then $u \in \mathcal{E}_0$, and the factorization gives $4\|Au\|_2^2 = \langle (\mathcal{L} - I)u, u \rangle = 0$. Thus $Au = 0$, which is

$$\partial_z \left(e^{|z|^2/4} u(z) \right) = 0.$$

Hence $u(z) = e^{-|z|^2/4} h(\bar{z})$ for an entire h . Since $u \in L^2$, its norm is exactly the integral in (2). This proves the claim, including $k = 0$. $\square$

Lemma 4 (A Landau level cannot vanish on a line). *If $v \in \mathcal{E}_k$ and $v(x) = 0$ for every $x \in \mathbb{R}$, then $v = 0$.*

Proof. Use Lemma 3. Writing $w = \bar{z}$, induction gives

$$(A^\dagger)^k \left(e^{-|z|^2/4} h(w) \right) = e^{-|z|^2/4} \left(\frac{z}{2} - \partial_w \right)^k h(w),$$

where z is held fixed while ∂_w acts. Therefore restriction to $z = w = x \in \mathbb{R}$ yields

$$T_k h(x) := \sum_{j=0}^k \binom{k}{j} \left(\frac{x}{2} \right)^{k-j} (-1)^j h^{(j)}(x) = 0. \quad (3)$$

The left side is entire, so the identity holds for complex x as well.

Write $h(x) = e^{x^2/4}p(x)$. The exponential generating function of the operators in (3) is

$$\begin{aligned} \sum_{k=0}^{\infty} \frac{t^k}{k!} T_k h(x) &= e^{tx/2} h(x-t) \\ &= e^{x^2/4} e^{t^2/4} p(x-t). \end{aligned}$$

Consequently

$$T_k h(x) = e^{x^2/4} P_k(D) p(x), \quad D = \frac{d}{dx},$$

where

$$P_k(s) = k! \sum_{r=0}^{\lfloor k/2 \rfloor} \frac{(-s)^{k-2r}}{4^r r! (k-2r)!} = 2^{-k} i^k H_k(is). \quad (4)$$

For $k = 0$, (3) immediately gives $h = 0$. Suppose $k \geq 1$. The physicists' Hermite polynomial H_k has k distinct real zeros $\xi_1, \dots, \xi_k$ [4]. Hence the roots of P_k are the distinct imaginary numbers $-i\xi_j$. Every entire solution of $P_k(D)p = 0$ has the form

$$p(w) = \sum_{j=1}^k c_j e^{-i\xi_j w}. \quad (5)$$

It remains to use the Fock condition, which is the decisive obstruction. For $z = x + iy$ and $w = \bar{z} = x - iy$,

$$\begin{aligned} e^{-|z|^2/2} |h(\bar{z})|^2 &= e^{-(x^2+y^2)/2} \left| e^{(x-iy)^2/4} p(x-iy) \right|^2 \\ &= e^{-y^2} |p(x-iy)|^2. \end{aligned}$$

For a fixed y , (5) is a finite exponential sum in x :

$$p(x-iy) = \sum_{j=1}^k c_j e^{-\xi_j y} e^{-i\xi_j x}.$$

Distinct frequencies are orthogonal in mean, so

$$\lim_{R \rightarrow \infty} \frac{1}{2R} \int_{-R}^R |p(x-iy)|^2 dx = \sum_{j=1}^k |c_j|^2 e^{-2\xi_j y}.$$

If one c_j is nonzero, the right side is positive and the integral over $x \in \mathbb{R}$ is infinite. Tonelli's theorem then contradicts (2). Thus all $c_j = 0$, whence $h = 0$ and $v = 0$. $\square$

Proof of the main theorem

Proof of Theorem 1. Fix $k \geq 0$. Choose $s \in [1, 2]$ from

$$1 + \frac{1}{2} = \frac{1}{p} + \frac{1}{s}.$$

Because φ_k is a polynomial times a Gaussian, it belongs to every $L^s(\mathbb{C})$. The twisted Young inequality gives

$$\|Q_k f\|_2 = \|f \times \varphi_k\|_2 \leq \|f\|_p \|\varphi_k\|_s < \infty.$$

Thus $Q_k f \in \mathcal{E}_k$. Polar coordinates and Fubini give, with a nonzero normalization constant c independent of x ,

$$Q_k f(x) = c \int_0^\infty (f \times \mu_r)(x) \varphi_k(r) r \, dr = 0, \quad x \in \mathbb{R}.$$

The integral is absolutely legitimate: translation preserves L^p , while the Gaussian factor belongs to the dual space. Lemma 4 now implies

$$Q_k f = 0 \quad \text{for every } k \geq 0.$$

For completeness, use the special-Hermite heat kernel. The Laguerre generating function gives

$$K_t(z) = \frac{1}{2\pi} \sum_{k=0}^\infty e^{-(2k+1)t} \varphi_k(z) = \frac{1}{4\pi \sinh t} \exp\left(-\frac{1}{4} \coth(t) |z|^2\right).$$

The series converges rapidly for $t > 0$. Hence, in tempered distributions and in L^p ,

$$e^{-t\mathcal{L}} f = f \times K_t = \frac{1}{2\pi} \sum_{k=0}^\infty e^{-(2k+1)t} Q_k f = 0.$$

Moreover, K_t is a Gaussian concentrating at the origin and $\int_{\mathbb{C}} K_t = 1/\cosh t \rightarrow 1$. Strong continuity of twisted translations therefore gives $f \times K_t \rightarrow f$ in $L^p(\mathbb{C})$ as $t \downarrow 0$ for every $1 \leq p < \infty$ (the usual approximate-identity proof applies verbatim). Therefore $f = 0$. $\square$

Proof of Corollary 2. Twisted convolution with radial measures is covariant under rotations. Applying Theorem 1 after a rotation proves injectivity for every line $e^{i\theta}\mathbb{R}$. Every Coxeter system Σ_N contains at least one such line, so vanishing for centers on Σ_N implies vanishing on that line and hence $f = 0$. $\square$

Verification and novelty check

The exact checker `code/check_hermite_axis_ode.py` verifies for levels $0 \leq k \leq 12$ that:

- coefficient extraction from $e^{t^2/4-ts}$ gives (4);
- $P_k(s) = 2^{-k} i^k H_k(is)$ exactly;
- direct normal ordering on a generic polynomial agrees with $T_k(e^{x^2/4}p) = e^{x^2/4} P_k(D)p$; and
- the computed Hermite zeros are real and distinct.

It was run with

```
conda run --no-capture-output -n sandbox python
code/check_hermite_axis_ode.py --max-k 12
```

and passed all levels. This finite computation is not used as proof; the generating-function and Hermite-zero argument above handles every k .

The bounded novelty search was performed on 26 August 2026. The run’s registry, solution, attempt, and proof-gap indexes were searched for the three relevant arXiv ids and for “odd Coxeter”, “twisted spherical means”, and “single line”; no prior packet was found. The local parsed arXiv corpus was searched for the title, arXiv:1103.4571, and the odd-line terminology. The only directly relevant hits were the source, the adjacent papers arXiv:1204.2773 and arXiv:1204.3076, and the later arXiv:1306.5658, which still describes real-cone injectivity as unsolved. Web searches restricted to arXiv used the same phrases together with “real axis”, “Landau level”, and “polyanalytic Fock”. They found background papers on polyanalytic Fock spaces but no paper claiming the single-line theorem or the odd-Coxeter conjecture. This is a bounded search, not a proof of novelty.

Limitations and human-review recommendation

The theorem is confined to $1 \leq p \leq 2$, exactly the range in which twisted Young places every spectral projection in L^2 . It concerns lines through the origin and does not settle the source’s broader questions about arbitrary real-analytic curves. As usual, the hypothesis is interpreted wherever the twisted spherical means are defined pointwise; the proof itself uses only the resulting polar-integral identities.

The packet is recommended for expert review as a candidate full solution. The highest-value checks are: (i) consistency of the twist convention with $\mathcal{L} = 4A^\dagger A + I$ and the assignment of $Q_k f$ to $\mathcal{E}_k$; (ii) the normal-order generating function in Lemma 4; and (iii) the L^p special-Hermite heat-kernel reconstruction at $p = 1$.

References

- [1] R. K. Srivastava, *Coxeter system of lines are sets of injectivity for the twisted spherical means on $\mathbb{C}$* , arXiv:1103.4571.
- [2] R. K. Srivastava, *Real analytic expansion of spectral projection and extension of Hecke–Bochner identity*, arXiv:1204.3076.
- [3] P. Ganguly and S. Thangavelu, *Theorems of Chernoff and Ingham for certain eigenfunction expansions*, arXiv:2011.09940.
- [4] G. Szegő, *Orthogonal Polynomials*, American Mathematical Society Colloquium Publications, Vol. 23, 4th ed., 1975.